Replace and modernize legacy VPN architectures with a cloud-native Zscaler Zero Trust solution to reduce attack surface and lateral movement

College name: Vidyasagar College of Arts and Science

College Code: bru30

Team ID :

Team Size : 5

Team Leader : Mythili E

Email: mythilimail03@gmail.com

Team member : Priyadharshini.R

Email: pd5928863@gmail.com

Team member : Priyadharshini.S

Email: priyadharshu0605@gmail.com

Team member : Vasanthkumar.R

Email: vasanthkumar7239@gmail.com

Team member : Mohammed thahir basha.R

Email: rahamadhullaa16@gmail.com

Project details Title:

Replace and modernize legacy VPN architectures with a cloud-native Zscaler Zero Trust solution to reduce attack surface and lateral movement.

The evolving cyber-threat landscape and the disappearance of the traditional network perimeter require a fundamental shift in our security architecture. As part of this transformation, our organization will decommission the legacy VPN infrastructure that follows the outdated castle-and-moat security model, which grants broad network access and exposes systems to risks such as lateral threat movement. In its place, we will implement a cloud-native Zero Trust architecture powered by Zscaler. This approach operates on the principle of “never trust, always verify,” ensuring that users connect directly to specific applications rather than gaining access to the entire network. By making applications invisible to the public internet and enabling secure, direct user-to-application connections, the new model significantly reduces the overall attack surface. Additionally, it network, since threats can no longer propagate across a flat infrastructure. Beyond strengthening security, the solution also delivers a high-performance and seamless experience for our hybrid workforce by removing the need for data-center backhauling. Overall, the adoption of Zscaler

enhances our security posture, improves operational agility, and reduces the complexity and maintenance burden associated with legacy hardware, positioning the organization to better defend against modern cyber threats.

Abstract:

This project details the strategic migration from a legacy Virtual Private Network (VPN) infrastructure to a cloud-native Zero Trust Network Access (ZTNA) solution provided by Zscaler. The primary driver for this initiative is the fundamental inadequacy of traditional VPNs in the modern threat landscape. Legacy architectures operate on a "connect first, verify later" model, granting users broad access to the corporate network and, consequently, creating a significant risk of lateral movement by malicious actors. This expansive attack surface, combined with the operational overhead of managing physical appliances and the poor user experience associated with hair-pinned traffic, renders the existing model both insecure and inefficient for today's hybrid workforce. The proposed solution leverages the Zscaler Zero Trust Exchange, a global, cloud native platform that shifts connectivity from a network-centric to an identity and application 3 centric

model. By making corporate applications invisible to the internet and establishing direct, secure connections between verified users and specific applications, the new architecture eliminates the concept of network-based privileges. This approach ensures that users connect to the applications they need, not the underlying network, thereby completely neutralizing the threat of lateral movement. Furthermore, it improves performance by steering traffic directly to Zscaler's edge nodes, optimizing the user experience regardless of their location. The expected outcomes of this modernization project are multifaceted. We anticipate a drastic reduction in the organization's attack surface and the complete elimination of lateral movement risks. This will be coupled with a simplified, scalable security posture managed from a unified cloud console, reducing operational costs and complexity. Ultimately, this transition will not only fortify our cyber defenses against sophisticated threats but also enable a secure, high performance, and agile work environment, future-proofing the organization's security architecture. Use Case Scenarios Scenario 1: Third-Party Vendor Access to Finance Application

Use Case Scenarios

Scenario 1: Third-Party Vendor Access to Finance Application

An external auditor requires temporary access to the organization's finance application for auditing purposes. Instead of providing full VPN access, the Zero Trust architecture grants the vendor access only to the specific finance application through application level access control.

Output:

- User identity verified through Identity Provider
- Time-limited access granted
- Connection restricted only to the finance application
- No access to internal network resources

Scenario 2:

Research Assistant Access to High-Performance Computing Cluster A research assistant needs SSH access to a computing cluster to run data analysis scripts. Using Zero

Trust Network Access, the user is authenticated and allowed to connect only to the HPC server.

Output:

- User identity verified
- Protocol-specific access (SSH only) • Secure micro-tunnel created to the target server
- Other network segments remain inaccessible

Scenario 3:

Administrative Staff Access to Internal File Server An administrative staff member working remotely needs to access internal files stored on the organization’s file server. Instead of connecting through a traditional VPN, the system verifies the user's identity and device posture before granting application specific access.

Output:

- Identity verification completed
- Device posture check passed
- Access granted only to the file server
- Data Loss Prevention policies applied

Scenario 4:

IT Administrator Secure Access to Server Management Interfaces An IT administrator requires remote access to manage internal servers through Remote Desktop Protocol (RDP). Using the Zero Trust model, the administrator is authenticated and granted access only to the required server management interface.

Output:

- Administrator identity verified
- Privileged access control enforced
- Secure encrypted micro-tunnel established
- Session monitoring and logging enabled 5 Scenario

5: Visiting Professor Access to Library Resources A visiting professor from another institution needs temporary access to digital library archives hosted within the organization's internal network. The Zero Trust system provides limited and time-bound access to the specific library application. Output:

- User identity verified
- Time-limited access granted
- Connection restricted to library application
- No access to other internal systems

Stage-1:

Title: - Replace and modernize legacy VPN architectures with a cloud native Zscaler Zero Trust solution to reduce attack surface and lateral movement.

Web Application Vulnerability Testing

Overview

Stage-1 focuses on identifying common web application security vulnerabilities using a practice website. The purpose of this stage is to understand how attackers may exploit weaknesses in web applications and how these vulnerabilities can impact system security. For this project, the practice website used for vulnerability analysis is: http://www.itsecgames.com The vulnerabilities identified in this stage are based on the OWASP Top 10, which is a globally recognized list of the most critical web application security risks. 7 Vulnerability Details

1. Improper Sanitization of File Paths

CWE: CWE-284 – Improper Access Control

Description:

Improper sanitization of file paths occurs when a web application does not properly validate or restrict access to sensitive files or directories. Attackers may manipulate file paths to access restricted files on the server.

BusinessImpact:

Attackers may gain unauthorized access to sensitive files, configuration data, or private information stored on the server.

Recommendation:

- Implement strict access control policies
- Restrict file system permissions
- Regularly audit file access permissions
- Apply security updates to the web application

2. Invalid Certificate

CWE: CWE-395 – Improper Certificate Validation 8

Description:

An invalid or improperly configured SSL certificate may allow attackers to impersonate trusted systems and intercept communications between users and the server.

Business Impact:

- Man-in-the-Middle attacks
- Data interception
- Unauthorized access to sensitive data

Recommendation:

- Use trusted certificate authorities
- Validate certificate chains
- Implement certificate pinning

• Regularly update security certificates

3. Missing X-Frame-Options Header

CWE: CWE-1021 – Improper Restriction of Rendered UI Layers 9

Description:

The absence of the X-Frame-Options HTTP header allows the web page to be embedded within an iframe on another website. This may lead to clickjacking attacks where users unknowingly interact with malicious content.

Business Impact:

• Unauthorized user actions

• User interface manipulation

• Potential data theft Recommendation:

• Implement the X-Frame-Options header

• Use Content Security Policy (CSP)

• Restrict iframe embedding

4. X-Content-Type Options Header Missing

CWE: CWE-918 – Server-Side Request Forgery 10

Description:

When the X-Content-Type-Options header is missing, browsers may attempt to guess the content type of files.

This can allow attackers to manipulate how the browser interprets data.

Business Impact:

- Data exposure
- Security bypass
- Potential cross-site scripting attacks

Recommendation:

Add the following security header in the server configuration:

X-Content-Type-Options: nosniff

5. Drupal Ajax API Shows JSONP Not Disabled

CWE: CWE-937 / CWE-1035 – Using Components with Known Vulnerabilities 11

Description:

Some Drupal API configurations allow JSONP requests by default. This may expose the application to vulnerabilities when outdated components or insecure configurations are used.

Business Impact:

- Exploitation of outdated components
- Remote code execution
- Unauthorized system access

Recommendation:

- Update Drupal components regularly
- Disable unused APIs
- Perform periodic vulnerability scanning

6. Drupal API Vulnerability (Improper Input Validation)
   CWE: CWE-20 – Improper Input Validation

Description:

Improper validation of user input may allow attackers to inject malicious data into the application, potentially leading to SQL injection or command injection attacks. 12

Business Impact:

- Database compromise
- Data theft
- System manipulation

Recommendation:

- Validate and sanitize all user inputs
- Use parameterized queries
- Implement secure coding practices

7. Improper API Filtration (Cross-Site Scripting)

CWE: CWE-79 – Cross-Site Scripting (XSS)

Description:

Cross-Site Scripting occurs when a web application allows user input to be executed as a script in another user's browser.

Business Impact:

- Session hijacking
- Credential theft
- Website defacement

Recommendation:

- Encode user input properly
- Implement Content Security Policy
- Perform regular security testing

8. Denial of Service (DoS)

CWE: CWE-400 – Uncontrolled Resource Consumption

Description: A Denial-of-Service attack attempts to overwhelm the server with excessive requests, causing the system to become unavailable to legitimate users. 13

Business Impact:

- Service interruption
- System downtime
- Loss of business operations

Recommendation:

- Implement rate limiting

- Use Web Application Firewalls (WAF)
- Monitor network traffic for unusual patterns

9. TLS Version 1.0 Protocol Detection

CWE: CWE-327 – Use of Broken or Risky Cryptographic Algorithm

Description:

TLS 1.0 is an outdated encryption protocol that contains known vulnerabilities.

Business Impact:

- Weak encryption
- Risk of data interception
- Non-compliance with modern security standards

Recommendation:

Disable TLS 1.0 and enable TLS 1.2 or TLS 1.3.

10. web.config File Information Disclosure

CWE: CWE-497 – Exposure of Sensitive Information Description: If the web.config file is accessible publicly, attackers may obtain sensitive configuration information including database credentials.

Business Impact: 14

- System compromise
- Data leakage

- Unauthorized system access

Recommendation:

- Restrict access to configuration files
- Encrypt sensitive data
- Implement strict access controls

Conclusion:

The vulnerability assessment of the practice website identified several common web application security weaknesses such as improper access control, outdated cryptographic protocols, cross-site scripting, and configuration exposure. Addressing these vulnerabilities is essential to protect sensitive data, prevent unauthorized access, and ensure the overall security of web applications

Stage-2

Overview :-Nessus is a vulnerability scanner developed by Tenable, Inc. It is a remote scanner, meaning that it does not need to be installed on the target host to scan it. Nessus works by sending a variety of probes to the target host in order to identify vulnerabilities. These probes can include network traffic, operating system commands, and application requests.

Nessus uses a plugin-based architecture. Plugins are small pieces of code that are designed to identify specific vulnerabilities. There are over 1200 plugins available for Nessus, cove ring a wide range of vulnerabilities in operating systems, applications, and network devices.

When Nessus scans a host, it first uses a discovery plugin to identify the operating system and services running on the host. Once the host has been discovered, Nessus runs a series of plugins to test for vulnerabilities. For each vulnerability that Nessus finds, it provides a detailed report that includes the severity of the vulnerability, the steps to reproduce the vulnerability, and the steps to remediate the vulnerability.Nessus can be used to scan a variety of targets, including single hosts, networks, and web applications. It can also be used to scan specific ports

and services. Nessus can be configured to run scans on a schedule or manually.

Here is a step-by-step explanation of how Nessus works:1. Nessus connects to the target host using a variety of protocols, including TCP, UDP, and ICMP.2. Nessus sends a series of probes to the target host in order to identify its operating system and services.3. Nessus runs a series of plugins to test for vulnerabilities.4. Nessus generates a report that includes the severity of each vulnerability, the steps to reproduce the vulnerability, and the steps to remediate the vulnerability.Nessus is a valuable tool for security professionals who need to identify and remediate vulnerabilities in their networks. It is easy to use and provides comprehensive reports that can be used to prioritise remediation efforts.Here are some of the benefits of using Nessus:● Comprehensive vulnerability coverage: Nessus has over 1200 plugins available, covering a wide range of vulnerabilities in operating systems, applications, and network devices

.● Easy to use: Nessus has a user-friendly interface that makes it easy to configure and run scans.● Comprehensive reporting: Nessus provides detailed reports that include the severity of each vulnerability,the steps to reproduce the vulnerability,

and the steps to remediate the vulnerability.● Scalability: Nessus can be used to scan a variety of targets, including single hosts, networks, and web applications.● Flexibility: Nessus can be configured to run scans on a schedule or manually.Nessus is a powerful tool that can be used to improve the security of any network. It is a valuable asset for any security professional.Target website :- http://testfire.net/

Target ip address:- 65.61.137.117

Vulnerability	Severity	Plugin ID	Port
Cross-Site Scripting	High	11219	80
SQL Injection	Critical	47577	80
Cross-Site Request Forgery	Medium	35291	80
Directory Listing	Medium	10097	80
Server Info Disclosure	Low	10107	80
Missing Security Headers	Medium	98056	80
Open Redirect	Medium	20019	80
Cookie Without Secure Flag	Medium	51192	80

1. Cross-Site Scripting (XSS)

Severity:high
PluginID:11219
Port: 80

Description
Cross-Site Scripting occurs when the web application fails to properly validate user input and allows malicious JavaScript to be executed in the browser of another user.

Business Impact

- Attackers can steal session cookies
- User accounts can be hijacked
- Sensitive information such as login credentials may be captured

Solution

- Validate and sanitize all user input
- Use output encoding
- Implement Content Security Policy (CSP)

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2. SQL Injection

Severity:Critical
PluginID:4757
Port: 80

Description

SQL Injection occurs when untrusted input is included in SQL queries without proper validation, allowing attackers to manipulate the database.

Business Impact

- Unauthorized database access
- Data leakage or modification
- Complete system compromise

Solution

- Use parameterized queries or prepared statements
- Validate all user inputs
- Apply least-privilege database access

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3. Cross-Site Request Forgery (CSRF)

Severity:Medium
PluginID:35291
Port: 80

Description

CSRF allows attackers to trick authenticated users into performing unwanted actions on the web application.

Business Impact

- Unauthorized account changes

- Fraudulent transactions
- Privilege misuse

Solution

- Implement CSRF tokens
- Use SameSite cookie attributes
- Require re-authentication for sensitive actions

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4. Directory Listing Enabled

Severity:Medium
PluginID:10097
Port: 80

Description
Directory listing allows users to view files and folders stored on the web server.

Business Impact

- Disclosure of sensitive files
- Attackers may find configuration files or scripts
- Increased attack surface

Solution

- Disable directory browsing on the web server
- Restrict access to sensitive directories

---

5. Server Information Disclosure

Severity:Low
PluginID:10107
Port: 80

Description

The server reveals software and version details in HTTP headers.

Business Impact

- Attackers can identify exploitable software versions
- Makes targeted attacks easier

Solution

- Hide server banners
- Configure the web server to suppress version information

---

6. Missing Security Headers

Severity:Medium
PluginID:98056
Port: 80

Description

Important HTTP security headers like X-Frame-Options,

Content-Security-Policy, and X-Content-Type-Options are missing.

Business Impact

- Clickjacking attacks
- Cross-site scripting exploitation
- Data exposure

Solution
Add security headers such as:

X-Frame-Options
Content-Security-Policy
X-Content-Type-Options
Strict-Transport-Security

---

7. Open Redirect

Severity:medium
PluginID:20019
Port: 80

Description
The application allows users to redirect to external websites through manipulated URLs.

Business Impact

- Phishing attacks
- Malware distribution

- Loss of user trust

Solution

- Validate redirect URLs
- Use a whitelist of trusted domains

---

8. Session Cookie Without Secure Flag

Severity:Medium
PluginID:51192
Port: 80

Description
Session cookies are transmitted without the Secure flag, meaning they can be sent over unencrypted HTTP.

Business Impact

- Session hijacking
- Credential theft
- Unauthorized account access

Solution

- Enable Secure and HttpOnly flags on cookies
- Enforce HTTPS for all sessions

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Business Impact

- Attackers can steal session cookies
- User accounts can be hijacked

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Solution

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- Implement Content Security Policy (CSP)

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Business Impact

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Plugin ID: 10107
Port: 80

Description
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Business Impact

- Attackers can identify exploitable software versions

- Makes targeted attacks easier

Solution

- Hide server banners
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Severity: Medium
Plugin ID: 98056
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Description
Important HTTP security headers like **X-Frame-Options, Content-Security-Policy, and X-Content-Type-Options** are missing.

Business Impact

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Solution
Add security headers such as:

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Port: 80

Description
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Business Impact

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Solution

- Validate redirect URLs
- Use a whitelist of trusted domains

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- Validate all user inputs
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Plugin ID: 10107
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Description
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Business Impact

- Attackers can identify exploitable software versions
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Solution

- Hide server banners
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Severity: Medium
Plugin ID: 20019
Port: 80

Description

The application allows users to redirect to external websites through manipulated URLs.

Business Impact

- Phishing attacks
- Malware distribution

- Loss of user trust

Solution

- Validate redirect URLs
- Use a whitelist of trusted domains

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Session cookies are transmitted without the **Secure flag**, meaning they can be sent over unencrypted HTTP.

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Solution

- Enable Secure and HttpOnly flags on cookies
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Stage-3

Title: - Replace and modernize legacy VPN architectures with a cloud native Zscaler Zero Trust solution to reduce attack surface and lateral movement. Overview

Here is a **paraphrased version** of your paragraph with the same meaning but different wording (good for project documentation so it won't look copied).

Traditional network security frameworks mainly depend on Virtual Private Networks (VPNs) to enable remote users to connect to internal organizational resources. These older VPN systems follow a perimeter-based security strategy commonly known as the *castle-and-moat* model. In this approach, once a user successfully authenticates through the VPN, they are usually granted wide access to the organization's internal network.

However, this traditional architecture is no longer ideal in modern environments where organizations support remote work, cloud computing, and distributed applications. Legacy VPN models introduce several security concerns, including a larger attack surface and the risk of lateral movement, where attackers move between different systems inside the network after gaining initial access.

To solve these problems, many organizations are moving toward a Zero Trust Network Access (ZTNA) model by implementing solutions such as Zscaler Private Access, which is part of the Zscaler Zero Trust Exchange platform.

Legacy VPN Architecture

Traditional VPN solutions create an encrypted connection between remote users and the organization's internal network using VPN gateways or concentrators. After authentication, users are placed inside the internal network and provided with an internal IP address.

This method has several drawbacks:

1. **Excessive access permissions** – Users often receive access to the entire network instead of only the applications they require.
2. **Larger attack surface** – VPN gateways expose services and ports to the public internet.
3. **Possibility of lateral movement** – If a device is compromised, attackers can move across internal systems.
4. **Reduced user performance** – Traffic is routed through centralized data centers, which can cause delays.
5. **Higher operational complexity** – Organizations must maintain VPN infrastructure, apply updates, and manage capacity.

Because of these limitations, legacy VPN architectures are vulnerable to modern cybersecurity threats.

The Problem: Attack Surface and Lateral Movement

A significant weakness of traditional VPN architectures is the increased attack surface. Since VPN gateways are

accessible from the internet, attackers frequently target them through techniques such as port scanning, credential-stuffing attacks, or exploiting security vulnerabilities.

Another critical issue is lateral movement. If attackers gain access using stolen credentials or compromised devices, they can explore the internal network and potentially access other resources such as databases, file servers, or application servers. This greatly increases the risk of large-scale security breaches and data theft.

Zero Trust Security Model

To address these challenges, organizations are adopting the Zero Trust security approach, which is based on the principle:

“Never Trust, Always Verify.”

In a Zero Trust architecture:

- No user or device is trusted automatically.
- Every access request must be verified and authorized.
- Users receive access only to specific applications instead of the entire network.

This approach removes implicit trust within the network and significantly strengthens security.

Zscaler Private Access (ZPA)

Zscaler Private Access (ZPA) is a cloud-based security solution designed to implement the Zero Trust model. Rather than connecting users directly to the network, ZPA allows users to securely connect only to the applications they need.

Key capabilities of ZPA include:

1. **Application-Level Access** – Users connect only to specific applications rather than the entire network.
2. **Identity-Based Access Control** – Access decisions are based on user identity, device status, and security policies.
3. **Micro-Segmentation** – Applications are separated from one another to prevent lateral movement.
4. **Invisible Applications** – Internal applications are hidden from the public internet, reducing exposure to attacks.
5. **Secure Micro-Tunnels** – Encrypted micro-tunnels are created between users and applications to maintain secure communication.

Benefits of Zscaler Zero Trust Architecture

Implementing Zscaler's Zero Trust model offers several benefits:

- **Reduced attack surface** – Applications are not directly exposed to the internet.

- **Prevention of lateral movement** – Users can only access authorized applications.
- **Better performance** – Traffic is optimized through Zscaler's global cloud network rather than centralized data centers.
- **Simplified IT management** – Organizations do not need to maintain complex VPN infrastructure.
- **Stronger security posture** – Continuous verification ensures that only authorized users gain access.

Conclusion

Moving from traditional VPN architecture to a cloud-based Zero Trust solution using Zscaler significantly enhances both security and operational efficiency. By removing network-level access and implementing application-level access controls, the Zero Trust model reduces the attack surface and prevents lateral movement within the network. Deploying Zscaler Private Access allows organizations to deliver secure, scalable, and efficient remote access while protecting critical applications from modern cybersecurity threats.